



THE ANNUAL NATIONAL LEVEL ROBOTICS COMPETITION, PCCOE&R, PUNE

ROBO RASHTRA '27

RISE OF ROBONATION, BHARAT LEADS THE WAY

ORBITAL CLASH

RULEBOOK

ONE CORE. TWO SYSTEMS. ONE SURVIVOR.





INTRODUCTION

ORBITAL CLASH IS A COMPETITIVE ROBOTICS CHALLENGE IN WHICH TWO MANUALLY OPERATED WIRELESS ROBOTS COMPETE TO CONTROL THE ENERGY CORE WHILE ATTEMPTING TO PUSH THE OPPOSING ROBOT OUTSIDE THE ARENA.

THE EVENT TESTS ROBOT CONTROL, MECHANICAL DESIGN, PRECISION AND STRATEGIC DECISION-MAKING UNDER TIME CONSTRAINTS. TEAMS MUST BALANCE OFFENSIVE AND DEFENSIVE STRATEGIES WHILE ATTEMPTING TO SECURE THE ENERGY CORE WITHIN THEIR DESIGNATED EXTRACTION ZONE.



PRIZE POOL



30,000

BOT SPECIFICATIONS



SIZE AND WEIGHT RESTRICTION

- LENGTH: 300 MM
- WIDTH: 300 MM
- HEIGHT: NO RESTRICTION
- MAXIMUM WEIGHT: 5 KG
- DIMENSIONAL TOLERANCE: ± 20 MM
- THE ROBOT MUST BE STABLE AND STAND ON ITS OWN AT THE BEGINNING OF THE MATCH.

CONTROL MECHANISM

- THE ROBOT MUST BE MANUALLY OPERATED USING A PERMITTED WIRED/WIRELESS CONTROL SYSTEM. ONLY ONE TEAM MEMBER MAY OPERATE THE ROBOT DURING A MATCH.
- PARTICIPANTS USING WIRELESS SYSTEMS ARE RESPONSIBLE FOR ENSURING RELIABLE OPERATION OF THEIR ROBOTS.

BOT SPECIFICATIONS



POWER SUPPLY

- THE ROBOT MUST USE AN ONBOARD POWER SOURCE DURING GAMEPLAY. EXTERNAL POWER SOURCES ARE NOT PERMITTED.
- THE MAXIMUM VOLTAGE IS 12.5 V. BATTERIES MUST BE SECURELY MOUNTED AND HANDLED SAFELY.

MOTORS

THERE IS NO RESTRICTION ON THE TYPE, MAKE, MODEL, OR NUMBER OF MOTORS USED IN THE ROBOT. TEAMS MAY USE ANY MOTOR CONFIGURATION SUITABLE FOR THEIR DESIGN, PROVIDED THAT THE ROBOT COMPLIES WITH ALL OTHER COMPETITION RULES AND THE MAXIMUM POWER SUPPLY VOLTAGE OF 12.5 V.



TECHNICAL INSPECTION

ALL ROBOTS WILL UNDERGO A TECHNICAL AND SAFETY INSPECTION BEFORE GAMEPLAY TO VERIFY THEIR DIMENSIONS, WEIGHT, POWER SUPPLY, WIRELESS CONTROL SYSTEM AND OVERALL RULE COMPLIANCE.

FAILURE TO MEET MANDATORY TECHNICAL OR SAFETY REQUIREMENTS MAY RESULT IN DISQUALIFICATION.

TEAM SPECIFICATION

- EACH TEAM MUST CONSIST OF 2-5 MEMBERS. ONE MEMBER MUST BE DESIGNATED AS THE ROBOT OPERATOR FOR EACH MATCH.
- ONLY PERMITTED TEAM MEMBERS MAY ENTER THE DESIGNATED COMPETITION AREA, AND ALL TEAM MEMBERS MUST FOLLOW THE INSTRUCTIONS OF THE EVENT OFFICIALS.

GAMEPLAY



STARTING POSITION

- BOTH ROBOTS WILL BEGIN IN THEIR RESPECTIVE STARTING ZONES, WHILE THE ENERGY CORE WILL BE PLACED AT ITS DESIGNATED STARTING POSITION.
- ROBOTS MUST REMAIN WITHIN THEIR STARTING ZONES UNTIL THE OFFICIAL START SIGNAL.

MATCH DURATION

- EACH MATCH WILL HAVE A MAXIMUM DURATION OF 3 MINUTES.
- THE TIMER WILL START WITH THE OFFICIAL START SIGNAL AND WILL CONTINUE THROUGHOUT GAMEPLAY.
- A TEAM MAY ABORT THE MATCH AT ANY TIME. IF A TEAM ABORTS, THE OPPOSING TEAM WILL BE DECLARED THE WINNER.

GAMEPLAY



QUALIFIER ROUND

- TEAMS WILL PRESENT THEIR ROBOTS TO THE JUDGES. ROBOTS WILL BE EVALUATED BASED ON DESIGN, FEATURES, FUNCTIONALITY AND IMPLEMENTATION.
- TEAMS WILL THEN PARTICIPATE IN THE DESIGNATED QUALIFIER MATCHES. MATCH PERFORMANCE AND OFFICIAL SCORING EVENTS WILL BE RECORDED.
- TOP 16 TEAMS WILL QUALIFY FOR THE TOURNAMENT ROUND. THE RANKING METHODOLOGY WILL BE DETERMINED BY THE ORGANIZING COMMITTEE.

TOURNAMENT ROUND

THE 16 QUALIFIED TEAMS WILL COMPETE IN THE TOURNAMENT ACCORDING TO THE FORMAT AND MATCH STRUCTURE ANNOUNCED BY THE ORGANIZING COMMITTEE.

GAMEPLAY



WINNER DECLARATION

- TEAMS WILL ADVANCE THROUGH THE TOURNAMENT ACCORDING TO THEIR OFFICIAL MATCH RESULTS AND THE ANNOUNCED TOURNAMENT FORMAT.
- THE FINAL WINNERS WILL BE DETERMINED BASED ON THE OFFICIAL COMPETITION RESULTS AND SCORING SYSTEM.
- THE DECISION OF THE JUDGES REGARDING GAMEPLAY, SCORING, PENALTIES, MATCH RESULTS AND RULE INTERPRETATION SHALL BE FINAL AND BINDING.

MANUAL INTERVENTION

NO MANUAL INTERVENTION IS PERMITTED ONCE GAMEPLAY BEGINS WITHOUT PERMISSION FROM THE EVENT OFFICIALS. UNAUTHORIZED INTERVENTION MAY RESULT IN A PENALTY OR DISQUALIFICATION.

TECHNICAL TIMEOUT

IF A ROBOT BECOMES NON-FUNCTIONAL DURING GAMEPLAY, THE TEAM MAY REQUEST A TECHNICAL TIMEOUT OF UP TO 1 MINUTE FOR REPAIR. IF THE ROBOT IS NOT RESTORED WITHIN THE ALLOTTED TIME, THE TEAM WILL BE DISQUALIFIED FROM THE MATCH AND THE OPPOSING TEAM WILL BE DECLARED THE WINNER WITH THE APPLICABLE BONUS POINTS.

MATCH RULES



1. THE ENERGY CORE WILL BEGIN AT ITS DESIGNATED CENTRAL POSITION. BOTH ROBOTS MAY PUSH AND MANOEUVRE THE CORE USING THEIR PERMITTED MECHANICAL STRUCTURES.
2. EACH TEAM WILL HAVE A DESIGNATED EXTRACTION ZONE FOR SECURING THE ENERGY CORE.
3. TEAMS MUST ENSURE THAT THEIR ROBOT IS FULLY READY AND PREPARED WHEN THEIR TEAM IS CALLED FOR THE MATCH.
4. ROBOTS MAY PHYSICALLY INTERACT WITH AND PUSH THE OPPOSING ROBOT DURING GAMEPLAY. SUCCESSFULLY PUSHING THE OPPOSING ROBOT COMPLETELY OUTSIDE THE OPERATIONAL ARENA WILL EARN THE APPLICABLE POINTS UNDER THE OFFICIAL SCORING SYSTEM. THE PUSH-OUT WILL BE CONSIDERED SUCCESSFUL ONLY WHEN THE OPPOSING ROBOT HAS COMPLETELY CROSSED OUTSIDE THE OPERATIONAL ARENA.
5. IF THE PUSHING ROBOT ALSO EXITS THE ARENA AS A DIRECT RESULT OF THE SAME INTERACTION, THE PUSH-OUT WILL REMAIN VALID.
6. HOWEVER, IF A ROBOT MOVES COMPLETELY OUTSIDE THE ARENA ON ITS OWN, WITHOUT BEING PUSHED BY THE OPPOSING ROBOT, THE OPPOSITION TEAM WILL RECEIVE THE POINT.

MATCH RULES



7. AT THE END OF THE 3-MINUTE MATCH, THE TEAM THAT HAS SUCCESSFULLY SECURED THE ENERGY CORE WITHIN ITS DESIGNATED EXTRACTION ZONE WILL RECEIVE THE APPLICABLE BONUS POINTS.
8. IF THE ENERGY CORE REMAINS IN NO MAN'S LAND OR HAS FALLEN OUTSIDE THE ARENA WHEN THE MATCH ENDS, NO TEAM WILL RECEIVE BONUS POINTS.
9. THE ENERGY CORE WILL HAVE THE SPECIFICATIONS AND DIMENSIONS AS DEFINED BY THE EVENT OFFICIALS.
10. IF THE ENERGY CORE FALLS COMPLETELY OUTSIDE THE OPERATIONAL ARENA DURING GAMEPLAY, THE EVENT OFFICIALS WILL RESET THE ROBOTS AND THE CORE TO THEIR DESIGNATED POSITIONS. THE MATCH TIMER WILL CONTINUE DURING THE RESET, AND NO ADDITIONAL TIME WILL BE PROVIDED.
11. DELIBERATELY PUSHING OR ATTEMPTING TO PUSH THE ENERGY CORE OUTSIDE THE OPERATIONAL ARENA IS PROHIBITED AND MAY RESULT IN A PENALTY.
12. DELIBERATE ATTEMPTS TO DAMAGE THE OPPOSING ROBOT, ENERGY CORE OR ARENA ARE ALSO PROHIBITED AND MAY RESULT IN A PENALTY OR DISQUALIFICATION.

GENERAL RULES



- ROBOTS MUST COMPLY WITH ALL TECHNICAL SPECIFICATIONS.
- ONLY APPROVED WIRED/WIRELESS CONTROL SYSTEMS MAY BE USED.
- ONLY ONE OPERATOR MAY CONTROL THE ROBOT DURING A MATCH.
- NO TEST PRACTICE OR TRIAL RUNS WILL BE ALLOWED ON THE OPERATIONAL ARENA.
- PUSHING THE OPPOSING ROBOT IS PERMITTED.
- DELIBERATE DAMAGE TO ROBOTS OR ARENA ELEMENTS IS PROHIBITED.
- JAMMERS AND INTENTIONAL SIGNAL INTERFERENCE ARE PROHIBITED.
- UNAUTHORIZED MANUAL INTERVENTION IS PROHIBITED.
- TEAMS MUST FOLLOW INSTRUCTIONS GIVEN BY EVENT OFFICIALS.
- UNSAFE OPERATION OR UNSPORTSMANLIKE CONDUCT MAY RESULT IN PENALTIES OR DISQUALIFICATION.
- JUDGES' DECISIONS ARE FINAL AND BINDING.
- IN CASE OF A TIE, A PREDEFINED TIE-BREAKER ROUND WILL BE CONDUCTED AS DIRECTED BY THE JUDGE.



IMPORTANT NOTE

THE ORGANIZING COMMITTEE RESERVES THE RIGHT TO MODIFY OR CLARIFY THE RULES WHEN NECESSARY FOR SAFETY, FAIRNESS, TECHNICAL FEASIBILITY OR COMPETITION MANAGEMENT.

ANY OFFICIAL CHANGES WILL BE COMMUNICATED TO PARTICIPATING TEAMS BEFORE THE AFFECTED ROUND.

PARTICIPANTS ARE RESPONSIBLE FOR KEEPING THEMSELVES INFORMED ABOUT OFFICIAL EVENT UPDATES.